Analysis of the Chaotic Regime in the Filtered Hénon Map

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***Abstract*—The insertion of finite impulse response (FIR) filters in the feedback loop of chaotic generators is a known strategy for bandwidth limitation in chaos-based communication systems (CBCS). However, the interplay between the nonlinearity of the generating map and the linearity of the filter may disrupt the chaotic regime, compromising the application. This work systematically characterizes, by means of numerical simulations, the conditions under which chaos is preserved in the two-dimensional Hénon map with internal FIR filters, considering four filter types (lowpass, highpass, bandpass, and bandstop) and seven design windows, with a dedicated sweep over the Kaiser window. The largest Lyapunov exponent is estimated from the tangent-map dynamics by propagating one tangent vector with periodic renormalization over an ensemble of initial conditions anchored at the equilibrium point of the expanded system, with a full-basis Modified Gram–Schmidt recomputation of marginal cells, and the protocol is validated against the standard Hénon map. Bootstrap confidence intervals on the exponent quantify the statistical confidence of the chaotic-coverage estimates. The implementation is delivered as an open-source Python package with JIT-compiled cores via Numba and dedicated data-versioning and automated-testing infrastructure. Results show that FIR filters do not eliminate the chaotic regime but rather reorganize it into geometrically intricate regions of the parameter space—interleaving chaotic, periodic, and divergent zones—whose shape depends systematically on the filter type and the design window employed.**

***Index Terms*—Chaos-based communication systems, Hénon map, FIR filters, Lyapunov exponent.**

I. Introduction

INFORMATION security in communication systems has traditionally been provided by encryption techniques at the upper layers of the reference model. In parallel, physical-layer security seeks to exploit the characteristics of the medium and of the transmitted signal itself in order to hinder interception by unauthorized receivers, without relying exclusively on upper-layer encryption [1]. In this context, chaos-based communication systems (CBCS) are among the approaches under study: by employing chaotic signals as information carriers, they exploit properties such as the wideband spectrum and the noise-like appearance of these signals [2].

A deterministic discrete-time chaotic signal has three characteristics that distinguish it from classical periodic or stochastic signals [3]: it is amplitude-bounded, aperiodic, and exhibits sensitive dependence on initial conditions (SDIC), that is, two orbits initially arbitrarily close diverge exponentially in phase space after a finite time [4]. These properties motivated, starting from the seminal work of Pecora and Carroll [5], the use of chaotic signals in telecommunications and signal processing. Pecora and Carroll [5] showed that two identical chaotic-signal generators can be synchronized, despite SDIC, when the receiver is driven by state variables of the transmitter. Such synchronization-based schemes fall within the broader class of *coherent* CBCS, as opposed to *non-coherent* architectures, which recover the transmitted information from statistical or structural properties of the received signal without reconstructing a synchronized replica at the receiver [6]. Since then, multiple CBCS applications have been proposed: modulation schemes [2], image coding and encryption [7], systems based on three-dimensional attractors [8], and, more recently, schemes with chaotic phase-locked loops [9].

A fundamental issue when applying chaotic signals to communication systems is the spectral width. Because transmission channels are always band-limited, it is necessary to know and *limit* the bandwidth of the transmitted chaotic signals—not to control it arbitrarily, but to confine it within the channel passband [10]. To this end, Eisencraft *et al.* [11] proposed the use of discrete-time linear filters embedded in the chaotic-signal generator, and Fontes *et al.* [12] consolidated the approach by showing that inserting the filter does not prevent chaotic synchronization. The behavior of the system under additional practical limitations, such as the digitalization of the numerical representation, was later investigated by Borges *et al.* [13].

A central question in this approach is whether the signals generated by the system with an inserted filter remain chaotic, given that the filter modifies the dynamics of the original generating map. This question has been investigated by Borges and Eisencraft [14], [15], who showed numerically, for low-order FIR filters, that the filter coefficients affect the behavior of the generated signals, with phenomena such as bifurcation cascades, coexistence of attractors, crises, and periodicity islands of the *shrimp* type [16]. Borges *et al.* [17] extended the

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analysis to FIR filters with a larger number of coefficients and different zero placements, and Borges *et al.* [18] to second-order all-pole filters. These works establish that the chaotic regime is not guaranteed to survive filtering and must be verified case by case—the starting point of the present study.

Whereas [17] characterized the filtered Hénon map across different zero-placement geometries of the FIR filter with a fixed design window, the present investigation treats *frequency selectivity* and *window design* as the primary axes. In addition to lowpass FIR filters, highpass, bandpass, and bandstop filters are considered, each designed through seven distinct windows, with a dedicated parametric sweep of the Kaiser shape parameter [19]. Through high-resolution dynamic classification maps in the filter order–cutoff frequency plane, we show that the preservation of chaos is governed, in a structured and only partly monotonic way, by the filter type and the design window: stopband attenuation correlates positively with chaotic coverage for lowpass, highpass, and bandpass filters, but the relation *reverses* for bandstop, where the low-attenuation rectangular window dominates. Aggregating the 124 sweeps further exposes a configuration-robust region of the order–cutoff plane in which chaos persists across windows and filter types. Relative to the previously published numerical characterizations of the filtered Hénon map by Borges *et al.* [14], [15], [17], which vary the zero-placement geometry of the FIR filter with a fixed design window, the present work adds two directions proper to computer engineering: (i) a statistical characterization of the chaotic coverage, with bootstrap confidence intervals on the Lyapunov exponent; and (ii) a description of the computational architecture of the implemented pipeline—a Python package with cores compiled by Numba [20], parallelization over the parameter grid, checkpoint versioning, and automated testing—as a condition for scientific reproducibility.

The objective of this work is to characterize systematically the conditions under which the chaotic regime is preserved in the Hénon map with an internal FIR filter, sweeping the order–cutoff frequency plane for four filter types and seven design windows. To this end, the largest Lyapunov exponent is estimated from the tangent-map dynamics by propagating one tangent vector with periodic renormalization, with a full-basis Modified Gram–Schmidt recomputation of marginal cells, and validated against the standard Hénon map; the resulting orbits are classified into high-resolution dynamic maps for the four filter types and seven windows; the effect of the Kaiser-window shape parameter β_K on the chaotic coverage is assessed; the chaotic-coverage estimates are accompanied by bootstrap confidence intervals on the exponent; and the whole pipeline is documented and released as a reproducible artifact [21]. This work covers the filter order, the cutoff frequency, the filter type, and the design window; other dimensions of the parameter space are left as future work enabled by the released infrastructure.

The remainder of this paper is organized as follows. Section II presents the theoretical background (Hénon map, expanded system with FIR filter, the frequency-domain rationale for feedback filtering, design windows, and Lyapunov exponent); Section III describes the experimental methodology, including the computational architecture of the pipeline; Section IV presents and discusses the results; and Section V summarizes the conclusions and proposes future work.

II. Theoretical Background

This section presents the theoretical tools used throughout the work, in the order in which they will be mobilized in the following sections. Subsection II-A defines the two-dimensional Hénon map, its expanded version under internal FIR filtering, and the corresponding Jacobian matrix. Subsection II-B motivates, from a master–slave synchronization standpoint, why the feedback filter must be specified and evaluated in the frequency domain. Subsection II-C describes the seven FIR design windows considered. Subsection II-D formalizes the Lyapunov exponent, the tangent-map recursion with Gram–Schmidt orthogonalization, and the adopted dynamic-classification criterion.

A. Hénon Map and Expanded System With FIR Filter

The two-dimensional Hénon map [4] is the discrete dynamical system adopted in this work as the chaotic generator. In its standard two-dimensional form, it is written as

$$\begin{cases} x_1(n+1) = \alpha - x_1^2(n) + \beta\, x_2(n), \\ x_2(n+1) = x_1(n), \end{cases} \tag{1}$$

where α and β are real parameters. For $\alpha = 1.4$ and $\beta = 0.3$, the map produces a strange attractor whose largest Lyapunov exponent is $\lambda_1 \approx 0.42$ and whose fractal dimension is ≈ 1.26 [3], [4], [15]. The Lyapunov exponent measures the average exponential rate at which two nearby orbits separate and thus quantifies the SDIC; it is formally defined in Subsection II-D, and for the present purposes it is enough to note that a positive λ_1 is the signature of chaos. From the dissipative relation $\det \mathbf{J} = -\beta$, with $|\beta| < 1$, follows the identity $\lambda_1 + \lambda_2 = \ln|\beta|$, valid for the two-dimensional system and used here as a consistency test for the implemented Lyapunov estimator (Subsection IV-A).

As proposed by Eisencraft *et al.* [11] and Fontes *et al.* [12], the bandwidth of the generated signals can be limited by filtering $x_1(n)$ through an FIR filter with N_s coefficients c_j, $0 \le j \le N_s - 1$, according to (2):

$$x_3(n) = \sum_{j=0}^{N_s-1} c_j\, x_1(n-j), \tag{2}$$

and feeding $x_3(n)$ back into the system in place of $x_1(n)$ in the nonlinear term. The filter order is $N_z = N_s - 1$, equal to the number of zeros of the transfer function [22]. Because the filter in (2) depends on the N_s most recent values of x_1, the instantaneous pair (x_1, x_2) is no longer a complete state of the system: the past samples $x_1(n-1), \ldots, x_1(n-N_s+1)$ must also be stored for the next iteration to be determined. To recover a first-order (memoryless) description, these delayed samples are kept as *auxiliary state variables* $x_4, \ldots, x_{N_s}$, each of which simply holds a one-step delay of the preceding variable. With this augmentation, for $N_s > 3$, the system is

written in the expanded state-space form of Borges *et al.* [14], [17], reproduced in (3):

$$\begin{cases} x_1(n+1) = \alpha - x_3^2(n) + \beta\, x_2(n), \\ x_2(n+1) = x_1(n), \\ x_3(n+1) = c_0\big(\alpha - x_3^2(n) + \beta\, x_2(n)\big) + c_1\, x_1(n) \\ \qquad\qquad +c_2\, x_2(n) + c_3\, x_4(n) + \cdots + c_{N_s-1}\, x_{N_s}(n), \\ x_4(n+1) = x_2(n), \\ x_j(n+1) = x_{j-1}(n), \quad j = 5, \ldots, N_s, \end{cases} \tag{3}$$

where the last two lines are precisely the delay chain that stores the filter memory. The dimension of the expanded system is $K = N_s$ (equal to the number of coefficients); for $N_s = 1$ (identity filter $c_0 = 1$) the original two-dimensional map (1) is recovered. A *fixed point* (also called an *equilibrium point*) of a discrete map $\mathbf{F}$ is a state $\mathbf{x}^\star$ that is left unchanged by one iteration, $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$; an orbit started exactly at $\mathbf{x}^\star$ stays there for all n [3]. For the two-dimensional map (1), solving $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$ gives two fixed points, of which the positive one is the relevant reference state. The same condition applied to the expanded system (3) defines its positive fixed point $\mathbf{p}^+$, which depends on the total gain $G = \sum_j c_j$ of the filter—equal to the filter gain at $\omega = 0$, i.e., its DC gain [22]—and can be obtained analytically following [17]; this property is used in the initialization of the ensemble of initial conditions (Subsection III-E).

The Jacobian matrix of system (3), evaluated at instant n, is given by [14], [17]

$$\mathbf{J}(n) = \begin{pmatrix} 0 & \beta & -2x_3(n) & 0 & \cdots & 0 \\ 1 & 0 & 0 & 0 & \cdots & 0 \\ c_1 & c_0\beta + c_2 & -2c_0 x_3(n) & c_3 & \cdots & c_{N_s-1} \\ 0 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & \cdots & 0 & 0 & 1 & 0 \end{pmatrix}, \tag{4}$$

of dimension $K \times K$ with $K = N_s$. Only the third row of $\mathbf{J}(n)$ depends explicitly on the filter coefficients $\{c_j\}$, showing that the entire interaction between the linear dynamics introduced by the filter and the quadratic nonlinearity of the map is concentrated in that state equation. This structure partially explains why the total gain $G = \sum_j c_j$ is a relevant scalar parameter of the dynamic behavior of the expanded system [17].

B. Frequency-Domain Rationale for Filtering the Feedback Loop

The reason the filter in (2) must be designed and evaluated in the frequency domain, rather than treated as an arbitrary linear combination of past samples, becomes clear once (1) is embedded in a coherent chaos-based communication architecture of the master–slave type [5]. Let $\mathbf{x}(n)$ and $\mathbf{y}(n)$ denote the state vectors of an autonomous transmitter (master) and a driven receiver (slave), respectively, and let $\mathbf{e}(n) \triangleq \mathbf{y}(n) - \mathbf{x}(n)$ be the synchronization error. When the nonlinearity acts on a single scalar carrier extracted from the state—as in (1), where that role is played by $x_1(n)$—the error dynamics reduce to the linear recursion

$$\mathbf{e}(n+1) = \mathbf{A}\, \mathbf{e}(n), \tag{5}$$

where $\mathbf{A}$ is the synchronization matrix; synchronization is attained whenever every eigenvalue $\lambda_k(\mathbf{A})$, $1 \le k \le K$, lies strictly inside the unit circle, $|\lambda_k(\mathbf{A})| < 1$ [23]. Under an ideal, distortion-free channel this condition is sufficient for the receiver to track the transmitter asymptotically and for the transmitted message to be recovered without degradation, apart from an initial transient.

Real transmission channels, however, are never distortion-free: physical limitations of the medium impose a finite passband, so the channel acts as a filter that reshapes the spectrum of the transmitted signal [10]. Because chaotic carriers spread their energy continuously over a wide band rather than concentrating it on a few spectral lines [2], even a modest truncation of that spectrum is enough to unsettle the coherent receiver: the driven (slave) system no longer evolves under the same effective nonlinearity as the master, the synchronization error ceases to vanish, and the recovered message estimate no longer tracks the original waveform, in either the time or the frequency domain [23]. Confining the transmitted spectrum to the channel passband *before* transmission is therefore not an optional refinement but a precondition for coherent detection over bandlimited channels [11], [12].

The FIR branch of (2) implements exactly this pre-shaping inside the feedback loop, and its two equivalent descriptions make the frequency-domain interpretation explicit. In the time domain, $x_3(n)$ is a finite convolution of $x_1(n)$ with the coefficients $\{c_j\}$; taking the z-transform of (2) under zero initial conditions yields the transfer function $H(z) = X_3(z)/X_1(z) = \sum_{j=0}^{N_s-1} c_j z^{-j}$, whose evaluation on the unit circle, $H(e^{j\omega})$, is the frequency response that fixes the passband, transition width, and stopband attenuation actually delivered by a given choice of $\{c_j\}$ [22]. It is this frequency response, not the raw coefficient values, that must be matched to the channel bandwidth—which is precisely why the filter order N_z and the cutoff frequency ω_c, rather than the individual c_j, are adopted as the sweep variables of the present study (Subsection III-C), and why the design method reviewed in Subsection II-C operates directly on that frequency response.

Embedding the filter in the feedback path, rather than only at the channel interface, has a further consequence for the synchronization matrix $\mathbf{A}$ itself, formalized next.

Lemma 1 (Synchronizability under FIR feedback filtering, adapted from [23])**.** *Let the master–slave pair satisfy the synchronization condition $|\lambda_k(\mathbf{A})| < 1$, $1 \le k \le K$. Inserting an FIR filter of order N_z in the feedback path, as in* (3)*, enlarges the synchronization matrix to dimension $(K+N_z)\times(K+N_z)$ by adding only null eigenvalues. Hence the original condition remains sufficient, independently of the coefficients $\{c_j\}$.*

By Lemma 1, synchronizability of the master–slave pair is preserved for *any* choice of coefficients $\{c_j\}$, provided the unfiltered pair already synchronizes—a guarantee that holds regardless of how aggressively the filter shapes the spectrum.

TABLE I
FIR WINDOWS EVALUATED AND THEIR TYPICAL STOPBAND ATTENUATIONS

Window	Atten. (dB)	Characteristic
Rectangular (boxcar)	21	Minimum reference
Bartlett	25	Triangular
Hamming	41	Standard reference
Hann	44	Raised cosine
Blackman	74	High attenuation
Blackman–Harris	92	Maximum attenuation
Kaiser (β_K variable)	variable	Parametric

Source: adapted from [24] and [22].

This result, however, concerns only the linear stability of the error recursion (5); it says nothing about whether the master's own trajectory $\mathbf{x}(n)$ remains chaotic once the same coefficients $\{c_j\}$ are fed back nonlinearly through (3) and enter the Jacobian in (4). Synchronization and chaoticity are thus two logically independent requirements: the former is guaranteed by construction, while the latter—as already anticipated in Section I and demonstrated numerically for low-order filters by Borges *et al.* [14], [15], [17]—is not. Consequently, a frequency-domain specification of $H(z)$ that merely fits the channel passband cannot, by itself, certify that the filtered generator remains a valid chaotic source; it must be complemented by the explicit dynamical verification developed in the remainder of this paper. This distinction motivates treating the window used to synthesize $H(z)$ as a first-class design variable, reviewed next.

C. FIR Design Windows

The FIR filters are designed by the method of windowing the ideal impulse response [22], [24], a technique that establishes a direct trade-off between stopband attenuation and transition-band width. Seven windows are investigated, as summarized in Table I. The attenuation values listed there are the peak side-lobe levels of each window's discrete-time Fourier transform, measured relative to the main-lobe peak; in the windowing method this side-lobe level fixes the minimum stopband attenuation that the resulting FIR filter can attain [22], [24]. The values grow from 21 dB for the rectangular window—whose abrupt truncation of the impulse response produces strong side lobes—to 92 dB for the four-term Blackman–Harris window, whose cosine-sum coefficients are chosen precisely to cancel side lobes [24]; this gain in attenuation is paid for with a wider main lobe and therefore a wider transition band.

The Hamming window is adopted as a historical reference because it was used in the original works of Fontes *et al.* [12] and Borges and Eisencraft [14] on the filtered Hénon map. The Kaiser window [19], parameterized by $\beta_K \geq 0$, allows the trade-off between attenuation and transition width to be varied continuously without changing the filter order—for $\beta_K = 0$ it degenerates into the rectangular window, and typical values $\beta_K \in [2, 14]$ yield windows whose attenuations correspond to those of the Bartlett, Hamming, Blackman, or Blackman–Harris windows [22].

The seven selected windows cover a broad spectrum of trade-offs between transition-band width and stopband attenuation: the rectangular window as the worst case, the Bartlett and Hann windows as classical intermediates, the Hamming window as the reference adopted in the previous filtered-Hénon literature [12], [14], and the Blackman and Blackman–Harris windows as representatives of high-attenuation windows—under the hypothesis that greater spectral attenuation favors the preservation of the chaotic regime by reducing side-lobe leakage into the rejected band. The Kaiser window, with β_K swept over 25 values, provides a continuous path connecting these regimes. The design of the coefficients $\{c_j\}$ is carried out by the `firwin` routine of the SciPy library [25].

D. Lyapunov Exponent and Dynamic Classification

By *sensitivity* here we mean the sensitive dependence on initial conditions (SDIC) introduced in Section I: how fast two orbits that start arbitrarily close to each other drift apart as the system evolves. This rate is made quantitative by the Lyapunov exponent, defined as the natural logarithm of the average factor by which the separation between two nearby orbits is multiplied at each iteration. An exponent equal to $\ln 2$ along a given coordinate means, for example, that the distance between the orbits doubles at every iteration in that direction, on average [3], [26]; a negative exponent means the orbits converge in that direction. For a discrete system of dimension K there are K such exponents, one per coordinate of the tangent space, and the SDIC of the generated signals is characterized by the largest of them, λ_1.

The tangent-map method is adopted [3], [26]. Consider a unit-radius hypersphere centered at the initial condition $\mathbf{x}(0)$, represented by a matrix $\mathbf{U}$ whose columns are orthonormal vectors. At each iteration, this hypersphere is transformed into an ellipsoid by the product of the Jacobian matrix $\mathbf{J}(n)$ of the expanded system, given by (4). To avoid numerical divergence or collapse of the axes along the iterations, Gram–Schmidt orthogonalization is applied at each step to the set $\{\mathbf{J}(n)\mathbf{w}_1(n), \ldots, \mathbf{J}(n)\mathbf{w}_K(n)\}$. The recursion of the orthonormal basis is therefore

$$\mathbf{y}_k(n+1) = \mathbf{J}(n)\,\mathbf{w}_k(n) - \sum_{j=1}^{k-1} \big\langle \mathbf{J}(n)\mathbf{w}_k(n),\, \mathbf{w}_j(n+1) \big\rangle\, \mathbf{w}_j(n+1), \tag{6}$$

$$\mathbf{w}_k(n+1) = \frac{\mathbf{y}_k(n+1)}{\|\mathbf{y}_k(n+1)\|}, \qquad 1 \leq k \leq K, \tag{7}$$

where the summation in (6) implements the Gram–Schmidt orthogonalization against the vectors already orthogonalized in the same iteration, and the normalization in (7) preserves the orthonormal character of the basis along the iterations; K is the dimension of the expanded system [14]. The Lyapunov exponent in the k-th direction is then

$$\lambda_k = \lim_{n\to\infty} \frac{\ln\|\mathbf{y}_k(1)\| + \cdots + \ln\|\mathbf{y}_k(n)\|}{n}. \tag{8}$$

The *orbit* of an initial condition $\mathbf{x}(0)$ is the sequence of states $\{\mathbf{x}(n)\}_{n\geq 0}$ produced by iterating the map; it is *bounded*

when there exists a finite M such that $\|\mathbf{x}(n)\| \leq M$ for all n, and *divergent* otherwise. The dynamic classification adopted for each point of the parameter grid is

$$\begin{cases} \lambda_1 > 0 & \rightarrow \text{chaotic}, \\ \lambda_1 \leq 0 & \rightarrow \text{periodic or stationary}, \end{cases} \tag{9}$$

conditioned on the boundedness of the orbit. The case $\lambda_1 \leq 0$ groups two asymptotically non-chaotic behaviors: convergence to an equilibrium point (fixed point) of the system—defined in Subsection II-A as the state $\mathbf{x}^\star$ satisfying $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$—and convergence to a periodic orbit [3]. Both are grouped into the "periodic or stationary" class, because the criterion of this work distinguishes only the presence ($\lambda_1 > 0$) or absence ($\lambda_1 \leq 0$) of chaos, and not the specific period of the limit orbit. First, however, it is necessary to separate the orbits that remain bounded from those that escape: the regime is classified as *divergent* when the state evolves to a non-finite value (NaN or infinity) at some iteration. In that case, λ_1 is discarded, since the boundedness hypothesis underlying (8) no longer holds.

The criteria and equations presented in this section underpin the experimental methodology described next, in which (8) is evaluated by numerical simulation over a dense grid of the filter parameter space.

III. Methodology

This section describes the experimental methodology, organized so that each design decision is introduced before it is used. Subsection III-A discusses the computational architecture that makes the parameter sweep tractable; Subsection III-B lists the filter types and windows evaluated; Subsection III-C specifies the sweep grid; Subsection III-D details the implementation of the Lyapunov estimator; Subsection III-E describes the ensemble of initial conditions and the global parameters; Subsection III-F defines the chaotic-coverage metrics; Subsection III-G introduces the statistical quantification; and Subsection III-H summarizes the reproducibility infrastructure.

The research adopts a quantitative-experimental approach. All the tooling was implemented in the Python package `chaotic_pfc`, released as open source under the BSD-3-Clause license [21], with sweep cores compiled via Numba [20] for parallel execution over the parameter grid. Each submodule referenced in this section corresponds to a module of the package, and the technical documentation in Sphinx is published via Read the Docs [21].

A. Computational Architecture of the Pipeline

Systems with SDIC impose characteristic computational constraints that deserve treatment within the scope of computer engineering. Reliable estimation of λ_1 requires long simulation horizons (hundreds to thousands of iterations per orbit) and an ensemble of initial conditions to mitigate sensitivity to transients [26]. Parametric sweeps in the $(N_z, \omega_c/\pi)$ plane, even at moderate resolution, generate thousands of independent points, each with its own ensemble: for a $40 \times 100 \times 25$ grid (filter order, cutoff frequency, initial conditions), a single sweep demands 10^5 evaluations of the Lyapunov estimator.

Three architectural decisions were adopted in the pipeline. First, Just-In-Time (JIT) compilation of the numerical cores via Numba [20], which translates annotated Python functions into machine code through LLVM, with a reduction of one to two orders of magnitude in execution time relative to the pure-Python fallback, as measured by the package benchmarks [21]. Second, explicit parallelization over the $(N_z, \omega_c/\pi)$ grid by means of `prange` loops, exploiting all the cores available on the host machine. Third, versioning of the sweep checkpoints as compressed NumPy files (`.npz`), allowing figures and tables to be recomputed without redoing the sweeps—a scientific-reproducibility requirement recommended for published computational artifacts.

The per-iteration cost of the production single-vector estimator is $\mathcal{O}(K)$, where $K = N_s = N_z + 1$ is the dimension of the expanded system. The full-spectrum kernel used to recompute the marginal cells and in the validation is more expensive, being dominated by the QR (Modified Gram–Schmidt) orthogonalization, $\mathcal{O}(K^3)$ per step, and by the Jacobian–basis product, $\mathcal{O}(K^2)$ per step. For N_{IC} initial conditions and up to N_{map} accumulation iterations, the full-spectrum cost per grid point is $\mathcal{O}(N_{\text{IC}} \cdot N_{\text{map}} \cdot K^3)$. For high orders ($N_z = 40$, $K = 41$), $N_{\text{map}} = 3{,}000$ and $N_{\text{IC}} = 25$, this estimate approaches 5×10^9 floating-point operations per grid point, making JIT compilation a practical requirement rather than a luxury.

B. Filter Types and Windows Evaluated

Four FIR filter types are investigated: lowpass, highpass, bandpass, and bandstop, all designed by the windowing method with the seven windows listed in Table I. For the band filters (bandpass and bandstop), the total width is fixed at $\Delta\omega = 0.05\,\pi$, distributed symmetrically around the center frequency ω_c. This value is a deliberate compromise. On the normalized axis $\omega/\pi \in (0, 1)$ it corresponds to a band of 5% of the available range, which spans about five steps of the cutoff grid (100 values over the open interval, Subsection III-C); the band is therefore narrow enough to keep the filter clearly frequency-selective—a genuine bandpass or bandstop rather than a near-allpass—yet wide enough to be resolved by the sweep and to admit a non-degenerate pass/stop band at every ω_c. Fixing $\Delta\omega$ also isolates the effect of the single geometric parameter under study, the center frequency ω_c; the band width is left as an independent variable for future work (Section V).

The design of the coefficients is carried out by `scipy.signal.firwin`, wrapped by the FIR-design routines of the package. For highpass and bandstop filters, the parity restriction of the windowing method requires N_z to be even [22], otherwise the filter ceases to have a null response at the desired frequency; this restriction is honored by the package when building the grid. For a small subset of the configurations (the Hann and Bartlett windows with lowpass and bandpass filters), the design routine returns non-finite coefficients for part of the grid; those cells are classified as divergent on account of the filter design, not the map dynamics.

C. Parameter Sweep Grid

For lowpass and bandpass filters, $N_z \in \{1, 2, \ldots, 40\}$ is swept with 100 values of normalized cutoff frequency ω_c/π linearly spaced in the open interval $(0, 1)$, totaling $4{,}000$ points per window–filter configuration. For highpass and bandstop filters, the parity restriction of the windowing method requires N_z to be even [22], reducing the set to $N_z \in \{2, 4, \ldots, 40\}$, with $2{,}000$ points per configuration. For the Kaiser window, an additional dedicated sweep covers $\beta_K \in [2.0, 14.0]$ with step 0.5 (25 values) for the four filter types, totaling 100 additional Kaiser configurations. Adding the combinations with fixed windows ($4 \times 6 = 24$), the experiment covers 124 sweeps.

D. Lyapunov Estimator: Implementation

Algorithm 1 summarizes the implementation of the Lyapunov estimator used to generate the results of this work, based on the tangent-map formulation of Wolf *et al.* [26] adapted to system (3). The production sweeps propagate a single tangent vector and renormalize it once every ten iterations, instead of at every iteration, which reduces the per-iteration cost from $\mathcal{O}(K^3)$ to $\mathcal{O}(K)$; the period between renormalizations follows the practice established since Benettin *et al.* [27], [28]. Cells whose single-vector estimate of λ_1 falls below a small margin (5×10^{-3}) are recomputed with the full-spectrum kernel, which reorthonormalizes the complete tangent basis with the Modified Gram–Schmidt algorithm, so that the classification near the transition boundary is not decided by the single-vector approximation. The Modified Gram–Schmidt algorithm is used because it is numerically more stable than the classical one when the tangent vectors become nearly parallel after many iterations.

The early-stop step substantially reduces the computational cost while preserving the adopted classification criterion: the criterion of *two* consecutive checkpoints mitigates premature stops in windows of chaotic intermittency (in which λ_1 oscillates for hundreds of iterations before stabilizing), and the full-spectrum recomputation of the marginal cells removes the classification risk introduced by the periodic renormalization. Recomputing the 372,000 cells of the study, 34,147 (9.18%) were handled by the full-spectrum kernel.

Algorithm 1 Single-vector Lyapunov estimation with periodic renormalization.

Input: initial condition $\mathbf{x}_0$; burn-in iterations N_{burn}; accumulation cap N_{map}; convergence tolerance ε.
$\mathbf{w} \leftarrow \mathbf{e}_1$; $s \leftarrow 0$; $\mathbf{x} \leftarrow \mathbf{x}_0$.
for $i = 1$ **to** N_{burn} **do**
  $\mathbf{x} \leftarrow \mathbf{F}(\mathbf{x})$ (discard transient).
end for
for $i = 1$ **to** N_{map} **do**
  $\mathbf{x} \leftarrow \mathbf{F}(\mathbf{x})$ (orbit evolution).
  if $\mathbf{x}$ is non-finite **then**
    return NAN (divergent orbit).
  end if
  $\mathbf{w} \leftarrow \mathbf{J}(\mathbf{x})\,\mathbf{w}$ (tangent-vector evolution).
  if i is a multiple of 10 **then**
    $n \leftarrow \|\mathbf{w}\|$; $s \mathrel{+}= \ln \max(n, 10^{-300})$; $\mathbf{w} \leftarrow \mathbf{w}/n$ (normalization).
    (early stop) every 100 iterations, if the variation of $\lambda_1 = s/i$ stayed below ε for two consecutive checkpoints and $i \geq 500$, terminate.
  end if
end for
if $|\lambda_1|$ lies below the marginal margin **then**
  recompute the cell with the full-spectrum kernel (Modified Gram–Schmidt on all columns).
end if
return s/i.

E. Ensemble of Initial Conditions and Global Parameters

For each grid point (N_z, ω_c), 25 initial conditions are sampled uniformly in a half-open cube of edge $\epsilon = 0.1$: each component of the initial state is displaced from the positive equilibrium point $\mathbf{p}^+$ of the expanded system by ϵU, with $U \sim \mathcal{U}[0, 1)$, so that it lies in $[p_k^+, p_k^+ + 0.1)$ componentwise. The equilibrium point is computed analytically from the total gain $G = \sum_j c_j$ [17]. The estimator of Subsection III-D is run with $N_{\text{burn}} = 500$ discard iterations before accumulation over $N_{\text{map}} = 3{,}000$ iterations. The value reported for the point is the mean of λ_1 over the ensemble, classified according to criterion (9).

The choice of $N_{\text{burn}} = 500$ stems from the estimate of the characteristic relaxation time to the attractor in the standard Hénon map ($\sim 1/\lambda_1 \approx 2.4$ iterations), with an ample margin to accommodate longer transients in configurations close to bifurcation boundaries. In production, $N_{\text{map}} = 3{,}000$ acts as a cap rather than as the horizon actually used: the adaptive criterion stops the accumulation once λ_1 stabilizes, and the effective horizon recorded per configuration has median between approximately 700 and $1{,}200$ iterations, with a floor of 700; the cap is reached in at most 0.06% of the cells, in only two of the 124 configurations.

In all experiments: $\alpha = 1.4$ and $\beta = 0.3$ (standard parameters of the Hénon map); $\beta_K = 5.0$ in the general Kaiser-window sweep; and the pseudo-random generator seed fixed at 42, for reproducibility.

F. Chaotic Coverage Metrics

Each grid point is labelled, by the criterion of Subsection II-D, as chaotic, periodic/stationary, or divergent. The divergence rule operates in three steps. First, for a single initial condition, the estimator returns a non-finite value if the state becomes non-finite at any iteration of the accumulation window. Second, a cell is marked divergent when at least one of the 25 initial conditions produces a non-finite state during the 500 burn-in iterations. Third, an initial condition that becomes non-finite after the burn-in is discarded from the average, so the value reported for the cell is the mean of λ_1 over the surviving initial conditions. Let N_{grid} be the total number of grid points, N_b the number of points whose orbit is bounded (chaotic or periodic, i.e., non-divergent), and N_c the number of chaotic points. Two complementary chaotic-coverage metrics

are then reported. The *conditional* fraction $\rho_c^{\text{fin}} = N_c/N_b$ is the share of chaotic points among the bounded ones; it measures the propensity to chaos once numerical stability is ensured, deliberately excluding the divergent region so that it is not artificially deflated. The *absolute* fraction $\rho_c^{\text{abs}} = N_c/N_{\text{grid}}$ is taken over the whole grid and measures the usable fraction of the design space. The two metrics answer different design questions: ρ_c^{fin} is relevant when the designer can adjust N_z and ω_c to reach a specific non-divergent operating point, whereas ρ_c^{abs} is relevant when a broad operating range with few divergent configurations is sought. The difference in grid size ($4{,}000$ points for lowpass and bandpass against $2{,}000$ for highpass and bandstop) follows from the parity restriction on N_z inherent to the windowing method [22]; the comparisons between filter types use fractions precisely to neutralize this difference in sample size.

G. Statistical Treatment of the Sweep Results

The statistical characterization operates at two distinct levels, which are kept separate to avoid ambiguity. At the *point* level, the value of λ_1 reported for each grid point (N_z, ω_c) is the mean over the ensemble of $N_{\text{IC}} = 25$ initial conditions (Subsection III-E), and the per-point ensemble dispersion is retained as metadata. At the *regional* level, the chaotic region of a given window–filter configuration is summarized by the mean, standard deviation, and a 95% confidence interval of λ_1 computed *over the set of chaotic grid points* of that configuration (sample size equal to N_c). It is this regional mean—not the per-point ensemble mean—whose confidence interval is reported in Tables II and III; consequently, the standard deviation σ_{λ_1} measures the spatial homogeneity of the chaotic region rather than estimation noise at a single point. The interval is obtained by nonparametric bootstrap [29] with $1{,}000$ resamples and a fixed seed (42), so that it quantifies the sampling variability of the regional mean over the chaotic set. The corresponding routines are in the module `chaotic_pfc/analysis/stats.py` of the released package [21].

H. Reproducibility Infrastructure

Each complete sweep generates a `.npz` file containing the tensor $h = \lambda_1(N_z, \omega_c)$ averaged over the ensemble, its per-cell standard deviation `h_desvio`, the coefficient orders and the coefficient bank (`coef`), the cutoff grid `wcorte`, the window and filter-type labels, the effective iteration count `n_iters_used`, and a `metadata` block with the simulation parameters (N_{burn}, N_{map}, number of initial conditions, α, β, seed, Kaiser β_K, the adaptive-stop settings, and the fallback count); the chaotic/periodic/divergent classification is derived from h. The files are versioned in `data/sweeps/`, allowing figures and tables to be recomputed without recomputing the sweeps. Automated tests (in `tests/`) verify, at each repository commit: the identity $\lambda_1 + \lambda_2 = \ln|\beta|$ in the canonical map, the accuracy of the λ_1 estimate against published reference values [26], [30], and the consistency of the designed filters (coefficient sum, zero placement). Continuous integration runs the tests at each code change.

All the infrastructure described here is available as open source under the BSD-3-Clause license [21], with a Git version tag and a `CITATION.cff` file following the FORCE11 recommendations for the citation of scientific software. The technical documentation in Sphinx is published on Read the Docs and contains the description of each module, with usage examples and the API reference.

IV. Results and Discussion

This section presents and discusses the results of the parametric sweeps conducted according to the methodology of Section III. Subsection IV-A validates the Lyapunov estimator against the standard, unfiltered Hénon map; Subsection IV-B consolidates the configurations of greatest chaotic coverage by filter type; Subsections IV-C and IV-D detail the classification maps and the aggregate density; Subsection IV-E addresses the effect of the Kaiser-window parameter β_K; Subsection IV-F reports the statistical significance of the estimates and the ranking of the combinations; and Subsection IV-G discusses the computational cost and reproducibility. Results original to this work are reported in direct prose; observations that reproduce previous findings of Borges and Eisencraft [14], [15], [17] are marked as "confirms" or "is consistent with"; statements from the general literature are marked with a reference to their source.

A. Estimator Validation

The estimation protocol was validated on the *standard* Hénon map—the original, unfiltered two-dimensional map of (1) with the classical parameters $\alpha = 1.4$, $\beta = 0.3$ (recovered by setting $N_s = 1$, $c_0 = 1$), often referred to as the *canonical* case—for which the literature establishes $\lambda_1 \approx 0.42$ [3], [15], [26]. The run with the parameters described in Section III (ensemble of 25 initial conditions, $N_{\text{burn}} = 500$, $N_{\text{map}} = 3{,}000$, seed 42, early stop disabled) reproduced $\bar{\lambda}_1 = 0.4182$, with a relative error below 10^{-2}. This level of accuracy is sufficient to distinguish the dynamic regimes defined in (9). In addition, the identity $\lambda_1 + \lambda_2 = \ln|\beta|$—a direct consequence of the dissipation of the map—was satisfied to machine precision in all unfiltered configurations tested, confirming the algorithmic correctness of the estimator independently of the published reference value. This validation was performed with the full-spectrum estimator, which propagates the complete tangent basis with Modified Gram–Schmidt orthogonalization, whereas the parameter sweeps of Section IV used the single-vector production path. In the subsequent sweeps, the early stop (Subsection III-D) is enabled to reduce the computational cost.

B. Dynamic Classification of the Parameter Space

Table II consolidates, for each filter type, the configuration that maximizes the absolute number of chaotic points N_c in the parametric sweep, considering both the fixed-coefficient windows and the Kaiser window with the β_K that maximizes N_c. As a comparative reference, the second configuration in decreasing order of N_c is also reported (a fixed window for lowpass, highpass, and bandpass; Kaiser for bandstop,

TABLE II
CONFIGURATION OF LARGEST N_c PER FILTER TYPE, WITH MEAN, STANDARD DEVIATION, AND CONFIDENCE INTERVAL OF λ_1

Filter	**Window**	N_c	ρ_c^{fin}	ρ_c^{abs}	$\bar{\lambda}_1$	σ_{λ_1}	95% CI of $\bar{\lambda}_1$
Lowpass	Kaiser (β_K=13.5)	979	77.6%	24.5%	0.0239	0.0231	[0.0223, 0.0254]
	Blackman–Harris	961	74.8%	24.0%	0.0211	0.0232	[0.0197, 0.0227]
Highpass	Kaiser (β_K=11.5)	149	8.1%	7.5%	0.0178	0.0263	[0.0138, 0.0219]
	Blackman–Harris	144	7.9%	7.2%	0.0190	0.0269	[0.0151, 0.0237]
Bandpass	Kaiser (β_K=13.5)	434	13.0%	10.8%	0.0286	0.0297	[0.0258, 0.0315]
	Blackman–Harris	418	12.3%	10.4%	0.0261	0.0301	[0.0235, 0.0293]
Bandstop	Rectangular	118	65.2%	5.9%	0.0709	0.0491	[0.0627, 0.0800]
	Kaiser (β_K=4.0)	58	39.2%	2.9%	0.0726	0.0336	[0.0626, 0.0798]

95% confidence intervals obtained by bootstrap [29]. Full per-configuration tables are regenerated from the versioned checkpoints by the released package [21].

in which the first position is occupied by the rectangular window).

The Kaiser window leads in three of the four filter types, both in ρ_c^{fin} and in ρ_c^{abs}. However, this comparison must be interpreted with caution, since each fixed window corresponds to a single coefficient configuration per grid point, whereas the Kaiser window was evaluated at 25 values of β_K, giving it a higher sampling density. Disregarding Kaiser, the Blackman–Harris window is the dominant reference for lowpass, highpass, and bandpass, which is consistent with its higher spectral attenuation (92 dB). Excluding Kaiser, over the six fixed windows ($n = 6$), the Spearman correlation between stopband attenuation and chaotic coverage, with its exact two-sided permutation p-value over the 720 permutations, is $\rho = +0.94$ ($p = 0.0167$) for lowpass, $+0.83$ ($p = 0.0583$) for highpass, $+0.94$ ($p = 0.0167$) for bandpass, and -0.37 ($p = 0.4972$) for bandstop; eight tests were performed in total.

The bandstop filter constitutes a particular case: the rectangular window reaches $\rho_c^{\text{fin}} = 65.2\%$, the highest conditional fraction among all filter types. The bandstop filter rejects only a narrow band ($\Delta\omega = 0.05\,\pi$) away from $\omega = 0$, so its DC gain $G = \sum_j c_j$ remains close to unity. In absolute terms, however, its coverage is modest ($\rho_c^{\text{abs}} = 5.9\%$) because a large share of the grid is divergent at higher orders, as detailed in Subsection IV-C.

The lowpass filter with the Kaiser window ($\beta_K = 13.5$) reaches the highest absolute coverage among the 124 window–filter combinations swept in this work, with $\rho_c^{\text{abs}} = 24.5\%$ against the 5.9% of the bandstop—a ratio of approximately $4.2\times$ in absolute coverage, or $8.3\times$ in the number of chaotic points (979 against 118).

C. Classification Maps

Figures 1–4 present the dynamic classification maps in the $(N_z, \omega_c/\pi)$ plane for the configuration of largest N_c of each filter type, colored according to criterion (9): red for chaotic, blue for periodic, and gray for divergent.

The dynamic reading of these maps reinforces, now across four filter types and seven windows, two observations previously reported by Borges *et al.* [15], [17]. First, the total filter gain $G = \sum_j c_j$ scales the positive fixed point $\mathbf{p}^+$ of the expanded system, corroborating the analytical dependence

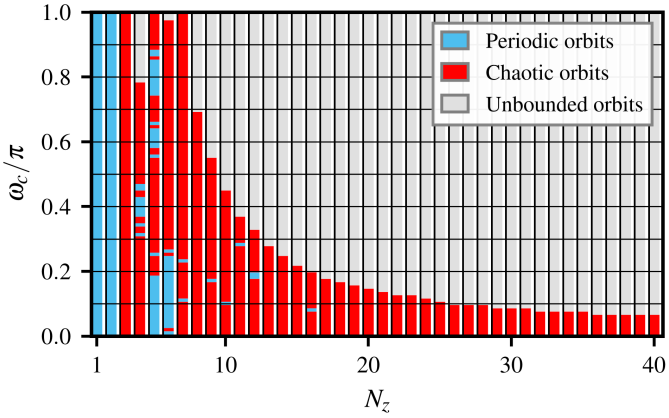


Fig. 1. Lowpass, Kaiser ($\beta_K = 13.5$). A wide chaotic region for $\omega_c/\pi \gtrsim 0.2$, with periodicity islands (*shrimps*) at low orders. The divergent region grows with N_z.

of $\mathbf{p}^+$ on G established in [17]. Second, the thin boundaries between chaotic and periodic regions are period-doubling bifurcation cascades, and the periodic islands inside the chaotic regions are *shrimps* [16]—periodic structures characteristic of two-dimensional quadratic maps, whose occurrence in the filtered Hénon map was first documented by Borges and Eisencraft [15]; our sweeps confirm that they persist across all four filter types studied here.

Figure 3 shows the map for the bandpass filter with the same Kaiser window ($\beta_K = 13.5$), in which a fragmented chaotic structure and greater sensitivity to variations of ω_c are observed, consistent with the narrow band ($\Delta\omega = 0.05\,\pi$) imposed by the filter.

Figure 4, finally, shows the map for the bandstop filter with the rectangular window. Its conditional coverage $\rho_c^{\text{fin}} = 65.2\%$ is the highest among all evaluated types, even though the absolute coverage is modest (5.9%) because of the high fraction of divergent points in the grid.

The four filter types differ substantially in the measured composition of the parameter grid. Pooling the 124 sweeps, the counts per type are: lowpass, 20,906 chaotic, 17,701 periodic, and 85,393 divergent cells out of 124,000 (68.87% divergent); highpass, 2,900, 54,902, and 4,198 out of 62,000 (6.77%); bandpass, 10,219, 100,990, and 12,791 out of 124,000 (10.32%); and bandstop, 1,598, 3,035, and 57,367

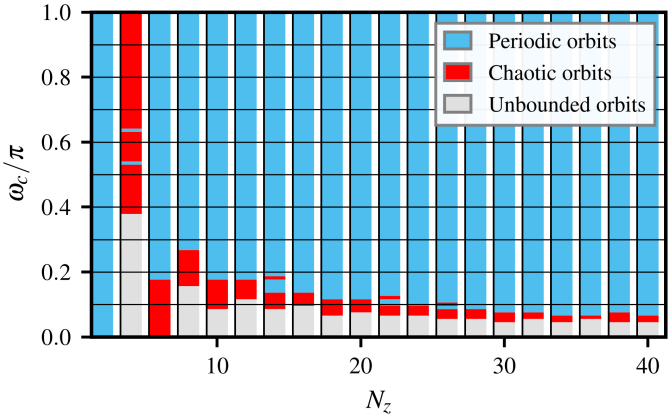


Fig. 2. Highpass, Kaiser ($\beta_K = 11.5$). Reduced chaotic coverage ($\rho_c^{\text{abs}} = 7.5\%$), concentrated at $\omega_c/\pi \lesssim 0.5$ and low orders, with the maximum at the lower edge of the swept interval—behavior opposite to that of the lowpass filter.

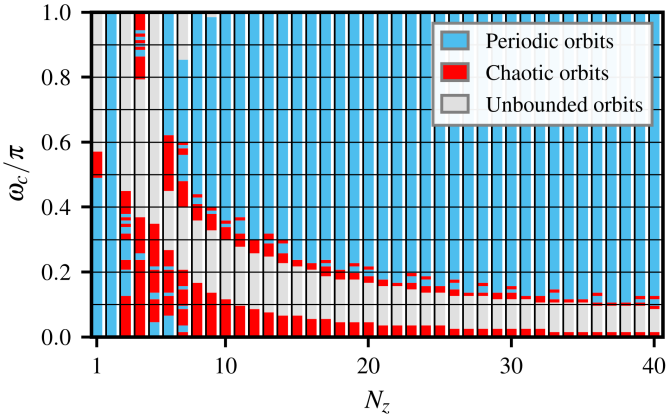


Fig. 3. Bandpass, Kaiser ($\beta_K = 13.5$), $\Delta\omega = 0.05\,\pi$. Fragmented chaotic structure and greater sensitivity to ω_c; maximum λ_1 equal to 0.196.

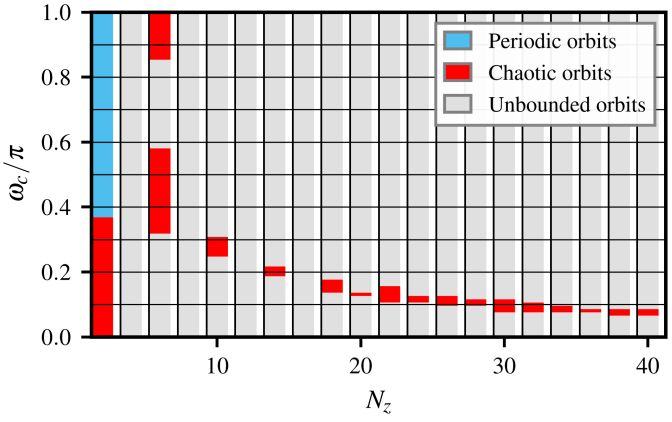


Fig. 4. Bandstop, rectangular window, $\Delta\omega = 0.05\,\pi$. Highest ρ_c^{fin} among all types (65.2%; $\rho_c^{\text{abs}} = 5.9\%$); absence of a divergent region at low orders.

out of 62,000 (92.53%), with cells classified as divergent under the rule of Subsection III-F. In the bandpass case, the chaotic set is limited predominantly by periodic orbits rather than by divergence (100,990 periodic against 12,791 divergent cells). The total gain $G = \sum_j c_j$ equals the filter frequency response at $\omega = 0$, i.e., its DC gain [22], and the positive

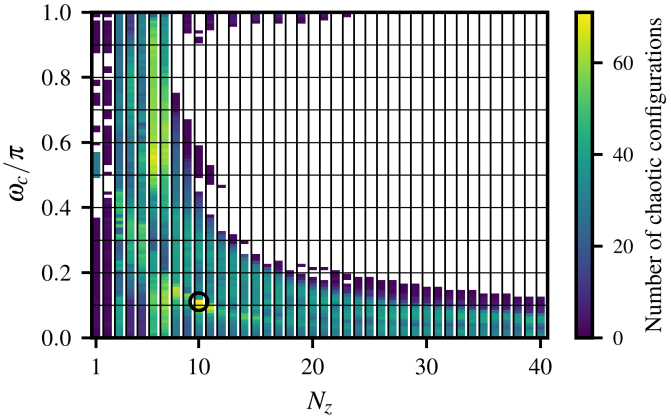


Fig. 5. Aggregate chaotic density over the 124 sweeps. Light tones indicate regions that are chaotic in a larger number of configurations; dark tones indicate fewer. The marker (○) highlights the point of maximum density, classified as chaotic in 71 of the 124 configurations.

equilibrium point $\mathbf{p}^+$ of the expanded system depends on G, as established analytically by Borges *et al.* [17] and recalled in Subsection II-A.

D. Aggregate Analysis and Chaotic Density

From the individual maps, all points classified as chaotic in each of the 124 sweeps (4 filter types × 6 fixed windows + 4 types × 25 values of β_K of the Kaiser window) were aggregated in the $(N_z, \omega_c/\pi)$ plane, regardless of the window or filter type. The result is the chaotic density map of Figure 5, which indicates which regions of the parameter space are classified as chaotic in the largest number of configurations.

The density map shows that the regions in which the chaotic regime is sustained across most of the windows and filter types are concentrated at intermediate-to-high cutoff frequencies and at moderate orders, regardless of the window or filter type. For high N_z, all configurations converge to similar dynamic behavior—a reflection of the approach to the ideal filter [22]—indicating that the choice of the window is a relevant design parameter mainly in the low-to-moderate order ranges.

The presence of *shrimps* [15], [16] and of bifurcation boundaries that are finely interleaved—formed by period-doubling cascades alternating with shrimp-shaped periodic islands at progressively smaller scales [15], [16]—implies that small variations of N_z or ω_c can move a configuration across a regime boundary, so the distance from each operating point to the nearest such boundary becomes an auxiliary design parameter.

E. Effect of the Kaiser-Window Parameter β_K

The Kaiser window is the only one of the seven evaluated that has a continuous shape parameter, β_K, providing a one-dimensional path through the space of windows [19]: for $\beta_K = 0$ it coincides with the rectangular window and, as β_K grows, the stopband attenuation increases at the expense of widening the main lobe [22]. This subsection examines how the chaotic coverage responds to this parameter, based on the dedicated

sweep described in Subsection III-B, in which β_K covers the interval $[2.0, 14.0]$ over 25 values for the four filter types.

The values of β_K that maximize the number of chaotic points N_c per filter type, reported in Table II, are $\beta_K = 13.5$ for lowpass and bandpass, $\beta_K = 11.5$ for highpass, and $\beta_K = 4.0$ for bandstop. For the first three types, the maximum lies at high values of β_K, where the window presents side lobes with strong attenuation [22]—behavior consistent with the finding of Subsection IV-B that windows of higher spectral attenuation, such as Blackman–Harris, favor the preservation of the chaotic regime in these three filter types. For lowpass in particular, it was observed that the absolute coverage ρ_c^{abs} grows sharply with β_K until it stabilizes at an approximate plateau for β_K between about 11 and 14, a range that contains the value $\beta_K = 13.5$ of largest N_c.

The distinct behavior of the bandstop filter, whose maximum N_c occurs with the rectangular window and, among the Kaiser configurations, at $\beta_K = 4.0$ (Table II), suggests a different interpretation: for filtering geometries that remove only a narrow band of the spectrum, the preservation of most of the spectral energy in the feedback loop seems to weigh more than the depth of the stopband attenuation—a hypothesis coherent with the best performance of the rectangular window, of lowest attenuation, in this same filter type. This difference reinforces that the effect of β_K is not monotonic across the four filter types, depending on the geometry of the imposed spectral mask.

F. Statistical Confidence and Configuration Ranking

The previous subsections reported, for each filter type, the configuration of greatest chaotic coverage. This subsection quantifies the statistical confidence of those estimates and consolidates the ordering of all window–filter combinations into a single reference table.

For each combination, the regional mean of λ_1 over the chaotic grid points (sample size N_c) is accompanied by a 95% confidence interval obtained by nonparametric bootstrap [29] with 1,000 resamples (seed 42), as defined in Subsection III-G. In all reported chaotic configurations, the confidence interval of $\bar{\lambda}_1$ lies entirely above zero—for example, $[0.0223, 0.0254]$ for lowpass with Kaiser ($\beta_K = 13.5$) and $[0.0627, 0.0800]$ for bandstop with the rectangular window (Table II)—confirming that the regional mean is positive with statistical confidence. The standard deviation σ_{λ_1}, being the spatial dispersion of λ_1 over the chaotic region, informs its homogeneity: the rectangular bandstop presents the highest value ($\sigma_{\lambda_1} = 0.0491$), reflecting the coexistence of low and high λ_1 in its fragmented chaotic region, while the lowpass with Kaiser exhibits a smaller dispersion ($\sigma_{\lambda_1} = 0.0231$) over a more homogeneous region.

The per-type trends are read directly from the classification maps and the per-type sweet spots: in lowpass filters the chaotic region concentrates at higher cutoff frequencies, whereas in highpass filters it migrates to low cutoff frequencies, with the highpass sweet spot located at $\omega_c/\pi \approx 0.01$ (Figs. 1 and 2). The per-configuration statistics are reproduced by the released package [21].

Table III consolidates the ranking of the window–filter configurations, considering the six fixed-coefficient windows and the best Kaiser window of each filter type, ordered by the number of chaotic points N_c.

The ranking in Table III shows that the first six positions are all of the lowpass type, confirming it as the filter of greatest absolute chaotic coverage, and that highpass and bandstop occupy mostly the lower half of the table. A relevant contrast for CBCS design is also observed: although the bandstop filter occupies low positions in N_c, its configurations present the highest values of $\bar{\lambda}_1$ (up to 0.0929), indicating that, when chaotic, their orbits exhibit greater sensitivity to initial conditions than those of the lowpass filter—a potentially desirable property depending on the application.

G. Computational Cost and Reproducibility

The complete sweep, comprising 124 window–filter combinations with an average grid of 3,000 points $\times$ 25 initial conditions, recorded per-cell effective accumulation horizons whose sum over the 372,000 cells is about 2×10^8, i.e., of order 5×10^9 iterations of the expanded map once the 25-initial-condition ensemble of each cell is counted. With cores compiled via Numba [20] and parallelization over the threads available on the machine used for the production runs (an 8th-generation Intel Core i7-8550U processor with 4 physical cores and 8 logical threads, 32 GB of RAM), the complete set of 124 sweeps took 489 s, about 8 min, measured end to end. Relative to an equivalent pure-Python implementation (without JIT), the package microbenchmarks during development indicated a reduction of one to two orders of magnitude in execution time—the reason for the architectural decision discussed in Subsection III-A.

All the infrastructure is versioned in the repository: sweep checkpoints in `data/sweeps/`, figure-generation scripts in `src/chaotic_pfc/analysis/sweep_plotting.py`, tests in `tests/`, and documentation in `docs/` (published on Read the Docs). The reproducibility of any figure or table of the present work can be obtained by running the corresponding script over the versioned checkpoint, without the need to recompute the sweep.

Taken together, the results of the previous subsections consolidate a characterization of the parameter space of the FIR-filtered Hénon map that is broader than the one documented in the previous literature; their implications are summarized in the following section.

V. Final Remarks

The dynamics of the Hénon map with FIR filters inserted in the feedback loop were investigated numerically, extending the results of Borges *et al.* [14] and [17], which vary the zero-placement geometry of the FIR filter with a fixed design window, to a higher-dimensional design space: four filter types (lowpass, highpass, bandpass, and bandstop) and seven windows (rectangular, Bartlett, Hamming, Hann, Blackman, Blackman–Harris, and Kaiser with a dedicated sweep of the shape parameter β_K).

TABLE III
RANKING OF THE WINDOW–FILTER COMBINATIONS (FIXED WINDOWS AND BEST KAISER PER FILTER TYPE), BY N_c

#	Filter	Window	N_c	ρ_c^{fin}	$\bar{\lambda}_1$	95% CI
1	Lowpass	Kaiser (β_K=13.5)	979	77.6%	0.0239	[0.0223, 0.0254]
2	Lowpass	Blackman–Harris	961	74.8%	0.0211	[0.0197, 0.0227]
3	Lowpass	Blackman	801	61.8%	0.0176	[0.0160, 0.0193]
4	Lowpass	Hann	606	51.2%	0.0190	[0.0170, 0.0208]
5	Lowpass	Bartlett	458	41.3%	0.0198	[0.0174, 0.0226]
6	Lowpass	Hamming	450	35.9%	0.0263	[0.0240, 0.0290]
7	Bandpass	Kaiser (β_K=13.5)	434	13.0%	0.0286	[0.0258, 0.0315]
8	Bandpass	Blackman–Harris	418	12.3%	0.0261	[0.0235, 0.0293]
9	Bandpass	Blackman	394	11.1%	0.0252	[0.0218, 0.0290]
10	Bandpass	Hamming	294	7.9%	0.0273	[0.0235, 0.0315]
11	Bandpass	Hann	288	8.0%	0.0267	[0.0231, 0.0311]
12	Bandpass	Bartlett	249	6.9%	0.0293	[0.0254, 0.0337]
13	Lowpass	Rectangular	204	19.5%	0.0354	[0.0314, 0.0396]
14	Highpass	Kaiser (β_K=11.5)	149	8.1%	0.0178	[0.0138, 0.0219]
15	Highpass	Blackman–Harris	144	7.9%	0.0190	[0.0151, 0.0237]
16	Bandpass	Rectangular	141	3.7%	0.0153	[0.0113, 0.0211]
17	Bandstop	Rectangular	118	65.2%	0.0709	[0.0627, 0.0800]
18	Highpass	Blackman	97	5.2%	0.0216	[0.0170, 0.0277]
19	Highpass	Bartlett	86	4.6%	0.0237	[0.0186, 0.0299]
20	Highpass	Hann	79	4.2%	0.0233	[0.0173, 0.0310]
21	Highpass	Hamming	72	3.8%	0.0234	[0.0175, 0.0300]
22	Bandstop	Kaiser (β_K=4.0)	58	39.2%	0.0726	[0.0626, 0.0798]
23	Bandstop	Hamming	55	35.5%	0.0874	[0.0854, 0.0892]
24	Bandstop	Blackman	50	33.3%	0.0888	[0.0865, 0.0912]
25	Bandstop	Hann	49	32.9%	0.0877	[0.0860, 0.0896]
26	Bandstop	Blackman–Harris	47	32.0%	0.0929	[0.0909, 0.0946]
27	Bandstop	Bartlett	43	30.1%	0.0896	[0.0882, 0.0911]
28	Highpass	Rectangular	25	1.3%	0.0269	[0.0171, 0.0398]

95% intervals by bootstrap [29]. Full per-configuration tables are regenerated from the versioned checkpoints by the released package [21].

It was confirmed that the insertion of the filters can substantially alter the nature of the generated signals, producing convergence to stationary regimes, periodic orbits, or chaotic behavior, depending on the adopted $(N_z, \omega_c/\pi)$ combination. The chaotic regime is not eliminated by the filtering, but rather reorganized into well-defined regions of the parameter space, dependent on the filter type and the window. For lowpass and bandpass filters, the Kaiser window with high β_K maximizes the chaotic coverage, reaching $\rho_c^{\text{abs}} = 24.5\%$ in the lowpass case—the highest absolute coverage among all configurations. For the bandstop filter, the rectangular window surpasses the others in conditional terms ($\rho_c^{\text{fin}} = 65.2\%$), although its absolute coverage is modest (5.9%).

The aggregate analysis over the 124 sweeps revealed that the regions in which the chaotic regime is sustained across most of the windows and filter types are concentrated at intermediate-to-high cutoff frequencies and moderate orders. The bootstrap confidence intervals on λ_1 place the regional mean of every reported chaotic configuration strictly above zero; read per filter type from the classification maps, the chaotic region migrates toward higher cutoff frequencies for the lowpass filter and toward lower ones for the highpass filter. The presence of *shrimps* [16] and of finely interleaved bifurcation boundaries reinforces the need for explicit verification of the generator dynamics after the insertion of linear filters—the preservation of chaos cannot be presumed.

Synthesis thesis. The joint choice of the filter type and the design window is a first-order parameter for the dynamic behavior of filtered Hénon generators: given the finely interleaved transition boundaries and the existence of *shrimps* inside the chaotic regions, the design of CBCS with filtering cannot start solely from the spectral specification of the filter; it requires explicit verification of the dynamics of the resulting generator by means of the classification maps produced here. The divergent fraction of the grid measured over the 124 sweeps ranges from 6.77% (highpass) and 10.32% (bandpass) to 68.87% (lowpass) and 92.53% (bandstop), with the counts reported in Subsection IV-C.

Acknowledged limitations. The sweeps cover $N_z \leq 40$ and horizons of $N_{\text{map}} \leq 3{,}000$ iterations; much higher orders and unbounded horizons could reveal additional structures. The band width $\Delta\omega$ was fixed at $0.05\,\pi$ for bandpass and bandstop; variations of this parameter may produce qualitative transitions in the conclusions for these two filter types. The choice of the canonical parameters $(\alpha, \beta) = (1.4, 0.3)$ of the underlying map was not swept; other pairs in the parameter space of the Hénon map [16] may present qualitatively distinct behaviors under filtering. The regional mean $\bar{\lambda}_1$ is computed only over the initial conditions that survive the accumulation window: initial conditions that diverge after the burn-in are discarded from the average, so λ_1 is estimated over the survivors.

Future work. Three directions are highlighted: (a) sweeping the band width $\Delta\omega$ as an independent variable for bandpass and bandstop filters, with the possibility of identifying additional transitions and dynamic resonances not captured in this study; (b) extending the parametric-sweep methodology to infinite-impulse-response (IIR) filters, whose dynamic behavior is still little explored in the literature [18], with explicit treatment of filter stability beyond map stability; and (c) a systematic spectral characterization of all the chaotic config-

urations identified, measuring occupied bandwidth, spectral leakage, and in-band power ratios as auxiliary metrics to λ_1 for the selection of operating points in CBCS.

REFERENCES

[1] A. Mukherjee, S. A. A. Fakoorian, J. Huang, and A. L. Swindlehurst, "Principles of physical layer security in multiuser wireless networks: A survey," *IEEE Communications Surveys & Tutorials*, vol. 16, no. 3, pp. 1550–1573, 2014.

[2] M. S. Baptista, "Chaos for communication," *Nonlinear Dynamics*, vol. 105, no. 2, pp. 1821–1841, 2021.

[3] K. T. Alligood, T. D. Sauer, and J. A. Yorke, *Chaos: An Introduction to Dynamical Systems*, ser. Textbooks in Mathematical Sciences. New York, NY, USA: Springer-Verlag New York, 1996.

[4] M. Hénon, "A two-dimensional mapping with a strange attractor," *Communications in Mathematical Physics*, vol. 50, no. 1, pp. 69–77, 1976.

[5] L. M. Pecora and T. L. Carroll, "Synchronization in chaotic systems," *Physical Review Letters*, vol. 64, no. 8, pp. 821–824, 1990.

[6] G. Kaddoum, "Wireless chaos-based communication systems: A comprehensive survey," *IEEE Access*, vol. 4, pp. 2621–2648, 2016.

[7] E. G. Sabatini, M. T. M. Silva, and M. Eisencraft, "Codificação de imagens usando sinais caóticos," in *Anais do XXXVIII Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, 2020.

[8] C. E. C. Souza, D. P. B. Chaves, and C. Pimentel, "Digital communication systems based on three-dimensional chaotic attractors," *IEEE Access*, vol. 7, pp. 10 523–10 532, 2019.

[9] R. V. Silva, D. P. B. Chaves, C. Pimentel, and M. S. Baptista, "Uma nova malha de travamento de fase aplicada a sistemas de comunicação baseados em atratores caóticos," in *Anais do XLIII Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Natal, RN, Brazil, 2025.

[10] M. Eisencraft, R. D. Fanganiello, and L. H. A. Monteiro, "Chaotic synchronization in discrete-time systems connected by bandlimited channels," *IEEE Communications Letters*, vol. 15, no. 6, pp. 671–673, 2011.

[11] M. Eisencraft, R. D. Fanganiello, and L. A. Baccalá, "Synchronization of discrete-time chaotic systems in bandlimited channels," *Mathematical Problems in Engineering*, vol. 2009, pp. 1–12, 2009.

[12] R. T. Fontes and M. Eisencraft, "A digital bandlimited chaos-based communication system," *Communications in Nonlinear Science and Numerical Simulation*, vol. 37, pp. 374–385, 2016.

[13] V. S. Borges and M. Eisencraft, "O efeito da digitalização num sistema de comunicação baseado em sincronismo caótico," in *Anais do XXXIX Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Virtual, 2021.

[14] ——, "Um estudo numérico do expoente de lyapunov do mapa de Hénon com filtro FIR," in *Anais do XL Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Sta. Rita do Sapucaí, MG, Brazil, 2022.

[15] ——, "A filtered Hénon map," *Chaos, Solitons & Fractals*, vol. 165, p. 112865, 2022.

[16] J. A. C. Gallas, "Structure of the parameter space of the Hénon map," *Physical Review Letters*, vol. 70, no. 18, pp. 2714–2717, 1993.

[17] V. S. Borges, M. T. M. Silva, and M. Eisencraft, "Chaotic properties of an FIR filtered Hénon map," *Communications in Nonlinear Science and Numerical Simulation*, vol. 131, p. 107845, 2024.

[18] ——, "Um estudo numérico da dinâmica do mapa de Hénon com filtros só-polos de segunda ordem," in *Anais do XLII Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Belém, PA, Brazil, 2024.

[19] J. F. Kaiser and R. W. Schafer, "On the use of the I_0-sinh window for spectrum analysis," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 1, pp. 105–107, 1980.

[20] S. K. Lam, A. Pitrou, and S. Seibert, "Numba: A LLVM-based Python JIT compiler," in *Proc. 2nd Workshop on the LLVM Compiler Infrastructure in HPC (LLVM '15)*, 2015, pp. 1–6.

[21] K. d. S. Silva, R. F. Pereira, M. F. C. Soares, G. R. d. Santos, and V. d. S. Borges, "chaotic-pfc: Parametric characterization of the filtered Hénon map," Software, BSD-3-Clause license, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.22740317

[22] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2009.

[23] V. d. S. Borges, "Dinâmica de sistemas de tempo discreto compostos por mapas não lineares e filtros lineares," Ph.D. dissertation, Escola Politécnica, Universidade de São Paulo, São Paulo, Brazil, 2026.

[24] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proceedings of the IEEE*, vol. 66, no. 1, pp. 51–83, 1978.

[25] P. Virtanen *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

[26] A. Wolf, J. B. Swift, H. L. Swinney, and J. A. Vastano, "Determining Lyapunov exponents from a time series," *Physica D: Nonlinear Phenomena*, vol. 16, no. 3, pp. 285–317, 1985.

[27] G. Benettin, L. Galgani, A. Giorgilli, and J.-M. Strelcyn, "Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems; a method for computing all of them. part 1: Theory," *Meccanica*, vol. 15, pp. 9–20, 1980.

[28] ——, "Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems; a method for computing all of them. part 2: Numerical application," *Meccanica*, vol. 15, pp. 21–30, 1980.

[29] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Boca Raton, FL, USA: Chapman and Hall/CRC, 1994.

[30] J. C. Sprott, *Chaos and Time-Series Analysis*. Oxford, UK: Oxford University Press, 2003.